

Input-to-state stabilization of linear systems under data-rate constraints

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Abstract—We study feedback stabilization of linear systems under data-rate constraints in the presence of completely unknown disturbances. A communication and control strategy is proposed based on sampled and quantized state measurements, where the quantization range is dynamically adjusted using reachable-set approximations and a disturbance estimate derived from quantization parameters. Under a data-rate condition, it guarantees input-to-state stability (ISS) with respect to the disturbance. An additional quantization symbol is introduced to establish ISS near the equilibrium. A simulation example illustrates the effectiveness of the proposed approach.

I. INTRODUCTION

Feedback control under data-rate constraints has been an active research area for decades, as surveyed in, e.g., [1]–[3]. Such constraints arise naturally in networked control systems due to communication costs, bandwidth limitations, and security considerations. Beyond these practical motivations, a fundamental question is how much information is required to achieve a given control objective.

A widely used framework for achieving finite data rate is to generate the control input from sampled and quantized state measurements taking values in a finite set. Beginning with [4]–[10], various quantization schemes have been developed for stabilizing linear systems. In particular, asymptotic stabilization under finite data rate requires dynamic adjustment of the size of the quantization range (zooming) [6]–[10], while the required data rate can be reduced by dynamically adjusting its center as well (moving-center quantization) [8]–[10]. These ideas have subsequently been extended to nonlinear systems [7], [11], [12] and switched systems [13], [14].

We consider feedback stabilization under data-rate constraints in the presence of unknown disturbances. In [8], [10], a known bound on the disturbance is assumed, and the minimum data rate required for asymptotic stabilization is characterized. Without such a bound, the problem becomes significantly more challenging, as disturbances may drive the state outside the quantization range after capture. In this setting, [15] established input-to-state stability (ISS) [16] using dynamic quantization with alternating zooming-out and zooming-in stages. Similar ISS results were obtained in [17] using logarithmic quantization in polar coordinates, and in [18] using moving-center quantization with improved data-rate efficiency. However, the fixed-center quantization

schemes in [15], [17] lead to implicit and generally conservative data-rate bounds, whereas the moving-center scheme in [18] relies on additional mechanisms such as a dedicated escape-detection mode and quantizer resets. More recently, [14] proposed a disturbance-estimation framework for switched linear systems that avoids these complexities, but achieves only the weaker property of practical ISS [19].

This paper addresses these limitations for linear systems with completely unknown disturbances. Extending the disturbance-estimation idea of [14], we propose a communication and control strategy that combines the data-rate efficiency of moving-center quantization with a Lyapunov-based design and eliminates the need for a dedicated escape-detection mode or quantizer resets. Our results provide an explicit characterization of the admissible data rate and establish ISS rather than practical ISS. To this end, we design a disturbance estimate derived from quantization parameters to replace the estimate based on an increasing sequence of constants in [14], and introduce an additional quantization symbol for establishing ISS near the equilibrium. These features distinguish the proposed approach from [14], [15], [17], [18] and constitute the main methodological contribution of the paper.

The remainder of the paper is organized as follows. Section II introduces the system model and information structure. Section III presents the main theorem. Section IV describes the communication and control strategy, while Section V derives the update formulas. Section VI provides the stability analysis. Section VII presents a simulation example. Section VIII concludes the paper.

Notation: The ∞ -norm of a vector $v = (v_1, \dots, v_n) \in \mathbb{R}^n$ is denoted by $|v| := |v|_\infty = \max_{1 \leq i \leq n} |v_i|$. The induced ∞ -norm of a matrix $M = [a_{ij}] \in \mathbb{R}^{n \times n}$ is denoted by $\|M\| := \|M\|_\infty = \max_{1 \leq i \leq n} \sum_{j=1}^n |a_{ij}|$. For a symmetric matrix $M \in \mathbb{R}^{n \times n}$, the largest and smallest eigenvalues are denoted by $\bar{\lambda}(M)$ and $\underline{\lambda}(M)$, respectively. A continuous function $\gamma : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is of class $\mathcal{K}_\infty$, denoted by $\gamma \in \mathcal{K}_\infty$, if it is strictly increasing, unbounded, and satisfies $\gamma(0) = 0$. The left limit of a piecewise continuous function $z(\cdot)$ at t is denoted by $z(t^-) := \lim_{s \nearrow t} z(s)$.

II. PROBLEM FORMULATION

A. System definition

Consider a continuous-time linear system

$$\dot{x} = Ax + Bu + Dd, \quad x(0) = x_0, \quad (1)$$

where $x \in \mathbb{R}^{n_x}$ is the state, $u \in \mathbb{R}^{n_u}$ is the control input, and $d \in \mathbb{R}^{n_d}$ is an unknown disturbance. The disturbance $d(\cdot)$ is assumed to be Lebesgue measurable and locally essentially

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bounded. The essential supremum ∞ -norm of $d(\cdot)$ over an interval J is denoted by $\|d\|_J := \text{ess sup}_{t \in J} |d(t)|$.

Our first basic assumption concerns stabilizability of (1).

Assumption 1 (Stabilizability). The pair (A, B) is stabilizable, that is, there exists a state feedback gain matrix K such that $A + BK$ is Hurwitz (all eigenvalues have negative real parts).

In what follows, we assume that such a matrix K has been selected and fixed.

B. Information structure

We aim to generate a stabilizing control input $u(\cdot)$ based on limited information about the state $x(\cdot)$.

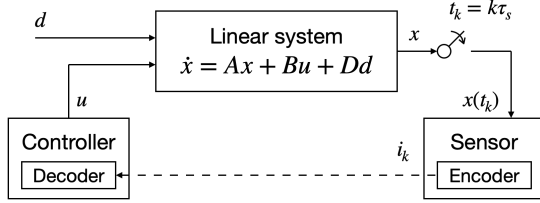


Fig. 1: Information structure.

As shown in Fig. 1, the feedback loop includes a sensor with an encoder and a controller with a decoder. The sensor samples the state at times $t_k = k\tau_s$, $k \in \mathbb{Z}_{\geq 0}$, where $\tau_s > 0$ is a fixed *sampling period*. Each sample $x(t_k)$ is encoded as an integer $i_k \in \{0, 1, \dots, N^{n_x} + 1\}$, where $N > 1$ is a fixed integer, and then transmitted to the controller. The resulting data transmission rate is

$$\frac{\log_2(N^{n_x} + 2)}{\tau_s}$$

bits per unit time. This information structure enables a separation of sensing and control tasks, since the controller does not require access to the exact state. Details of the communication and control strategy are presented in Section IV.

Our second basic assumption concerns the data rate.

Assumption 2 (Data rate). The sampling period τ_s and the integer N satisfy

$$\Lambda := \|e^{A\tau_s}\| < N. \quad (2)$$

The inequality in (2) requires N to be sufficiently large relative to τ_s , thereby imposing a lower bound on the admissible data rate. Similar data-rate conditions were shown to be sufficient for stabilizing linear systems [9], [18] and switched linear systems [13], [14]. These conditions, including (2), are generally conservative compared to the minimal data rate required for stabilization characterized in [8], [10]. However, the gap can be made arbitrarily small by tailoring the quantization scheme to the structure of the system matrix A ; see, e.g., [18, Sec. V]. Furthermore, the results in [8], [10] assume a known bound on the disturbance, whereas the disturbance is completely unknown in [14], [18] and in the present work.

III. MAIN RESULT

The control objective is to stabilize the system defined in Section II-A under the information structure described in Section II-B in a robust sense. Specifically, we aim to establish an *input-to-state stability* (ISS) [16] property with respect to the disturbance. The theorem below adopts a characterization of ISS from [19]; see also [15, Sec. II].

Theorem 1. Consider the linear system (1). Suppose that Assumptions 1 and 2 hold. Then there exists a communication and control strategy that yields the following ISS property with respect to the disturbance: there exist functions $\gamma_1, \gamma_2, \gamma_3 \in \mathcal{K}_\infty$ such that, for any initial condition $x_0 \in \mathbb{R}^{n_x}$ and any disturbance $d(\cdot)$, the closed-loop solution satisfies

$$|x(t)| \leq \gamma_1(|x_0|) + \gamma_2(\|d\|_{[0, \infty)}) \quad \forall t \geq 0 \quad (3)$$

and

$$\limsup_{t \rightarrow \infty} |x(t)| \leq \gamma_3\left(\limsup_{t \rightarrow \infty} |d(t)|\right). \quad (4)$$

As in [19], the limit supremum in (4) is understood in the essential sense. The communication and control strategy is presented in Section IV. The closed-loop dynamics are derived in Sections IV and V; see (12) and (24). The gain functions γ_1 , γ_2 , and γ_3 are derived in the proof of Theorem 1 in Section VI; see (42) and (43). In particular, γ_1 and γ_2 may depend on the choice of initial quantization parameters, but not on x_0 or $d(\cdot)$.

IV. COMMUNICATION AND CONTROL STRATEGY

In this section, we present the communication and control strategy used to establish Theorem 1, assuming that suitable approximations of the reachable sets of the state are available at all sampling times (their constructions are deferred to Section V).

The initial state $x(0) = x_0$ is unknown. At $t_0 = 0$, both the sensor and the controller are initialized with $x_0^* = 0$ and an arbitrary design parameter $E_0 > 0$. At each sampling time $t_k \geq 0$, the sensor checks whether

$$|x(t_k) - x_k^*| \leq E_k, \quad (5)$$

that is, whether the state $x(t_k)$ lies inside the hypercube

$$\mathcal{R}_k := \{v \in \mathbb{R}^{n_x} : |v - x_k^*| \leq E_k\}.$$

The set $\mathcal{R}_k$ approximates the reachable set of the state at t_k and serves as the quantization range. If (5) holds, then the state is *visible* and the system is in a *stabilizing stage* (Section IV-A); otherwise, the state is *lost* and the system is in a *searching stage* (Section IV-B).

If the state is visible at t_k , then the system remains in a stabilizing stage until the first sampling time $t_j > t_k$ such that $x(t_j) \notin \mathcal{R}_j$, at which point the state *escapes*. Conversely, if the state is lost at t_k , then the system remains in a searching stage until the first sampling time $t_i > t_k$ such that $x(t_i) \in \mathcal{R}_i$, at which point the state is *(re)captured*. Due to the unknown disturbance, the system may alternate between stages finitely or infinitely many times.

A. Stabilizing stages

At each sampling time t_k in a stabilizing stage, the sensor first checks whether

$$|x(t_k)| \leq \frac{E_k}{N}. \quad (6)$$

If so, it transmits $i_k = 1$ to the controller. Otherwise, it partitions the range $\mathcal{R}_k$ into N^{n_x} equal hypercubic cells of radius E_k/N in the ∞ -norm (N per dimension), assigns each cell a unique index from $\{2, \dots, N^{n_x} + 1\}$, and transmits the index i_k of the cell containing $x(t_k)$.

Upon receiving an $i_k \in \{1, \dots, N^{n_x} + 1\}$, the controller infers that (5) holds and reconstructs the corresponding cell center c_k using the same indexing protocol. If $i_k \geq 2$, then

$$|x(t_k) - c_k| \leq \frac{E_k}{N} \quad (7)$$

and

$$|c_k - x_k^*| \leq \frac{N-1}{N} E_k. \quad (8)$$

If $i_k = 1$, then $c_k = 0$ and (7) still holds.

The controller then applies the control input

$$u(t) = K\hat{x}(t), \quad t \in [t_k, t_{k+1}), \quad (9)$$

where K is the gain matrix from Assumption 1, and $\hat{x}$ evolves according to the auxiliary system

$$\dot{\hat{x}} = A\hat{x} + Bu \quad (10)$$

with the boundary condition

$$\hat{x}(t_k) = c_k. \quad (11)$$

That is, $\hat{x}$ is updated to c_k at each sampling time t_k and is therefore only right-continuous in general. Consequently, the controller dynamics are hybrid, comprising continuous flows and discrete jumps.

Finally, both the sensor and the controller update the quantization range by computing

$$x_{k+1}^* := F(c_k), \quad E_{k+1} := G(E_k, x_k^*)$$

without further communication, where the functions F and G are derived in Section V-A.

In contrast to the quantization schemes in [9], [13], [14], [18], our scheme includes an additional quantization cell centered at the origin, corresponding to (6). This design ensures that the control input vanishes whenever the state is sufficiently close to the origin, which is essential for establishing ISS. It also preserves the equilibrium at the origin without requiring the integer N to be odd.

B. Searching stages

At each sampling time t_k in a searching stage, the sensor transmits the “overflow symbol” $i_k = 0$. Upon receiving $i_k = 0$, the controller infers that the state is lost and sets the control input to zero on $[t_k, t_{k+1})$. Finally, both the sensor and the controller update the quantization range by computing

$$x_{k+1}^* := \hat{F}(x_k^*), \quad E_{k+1} := \hat{G}((1 + \varepsilon)E_k, \delta)$$

without further communication, where the functions $\hat{F}$ and $\hat{G}$ are derived in Section V-B, and $\varepsilon, \delta > 0$ are arbitrary design parameters. At an escape time t_j (i.e., the start of a searching stage), the update of E_{j+1} is modified as described in Section V-B.

V. APPROXIMATION OF REACHABLE SETS

In this section, we derive the update formulas used in the communication and control strategy presented in Section IV by constructing suitable approximations of reachable sets.

A. Stabilizing stages

Consider a sampling time t_k in a stabilizing stage. Define the error $e := x - \hat{x}$. On $[t_k, t_{k+1})$, combining (1), (9), and (10) yields the closed-loop system

$$\dot{x} = Ax + BK\hat{x} + Dd, \quad \dot{\hat{x}} = (A + BK)\hat{x}, \quad (12)$$

with the boundary condition (11), and thus the error dynamics

$$\dot{e} = Ae + Dd, \quad |e(t_k)| = |x(t_k) - c_k| \leq \frac{E_k}{N}, \quad (13)$$

where the inequality follows from (7). Hence

$$\begin{aligned} |e(t_{k+1}^-)| &\leq \|e^{A\tau_s}\| |e(t_k)| + \left(\int_0^{\tau_s} \|e^{As} D\| ds \right) \|d\|_{[t_k, t_{k+1}]} \\ &\leq \frac{\Lambda}{N} E_k + \Phi \|d\|_{[t_k, t_{k+1}]}, \end{aligned} \quad (14)$$

where $\Lambda = \|e^{A\tau_s}\|$ is defined in (2), and

$$\Phi := \int_0^{\tau_s} \|e^{As} D\| ds. \quad (15)$$

We define the update formulas by

$$x_{k+1}^* = F(c_k) := \hat{x}(t_{k+1}^-) = Sc_k \quad (16)$$

with $S := e^{(A+BK)\tau_s}$, and

$$E_{k+1} = G(E_k, x_k^*) := \frac{\Lambda}{N} E_k + \phi \sqrt{V_k} \quad (17)$$

with

$$V_k := V(x_k^*, E_k) := (x_k^*)^\top P x_k^* + \rho E_k^2, \quad (18)$$

where $\phi, \rho > 0$ are design parameters selected below. The functions F and G are designed such that, if the disturbance $\|d\|_{[t_k, t_{k+1}]} = 0$, then $|x(t_{k+1}) - x_{k+1}^*| \leq E_{k+1}$, so the state cannot escape at the next sampling time t_{k+1} . The term $\phi \sqrt{V_k}$ in (17) ensures that, if the state does escape at t_{k+1} , then V_k can be bounded in terms of $\|d\|_{[t_k, t_{k+1}]}$.

We now select three design parameters $\psi, \rho, \phi > 0$ sequentially so that V_k is a Lyapunov-type function with exponential decay during stabilizing stages. Since $A + BK$ is Hurwitz, there exist positive definite symmetric matrices $P, Q \in \mathbb{R}^{n_x \times n_x}$ such that

$$S^\top P S - P = -Q < 0. \quad (19)$$

Define

$$\chi := \frac{2n_x^2 \|S^\top P S\|^2}{\lambda(Q)} + n_x \|S^\top P S\|. \quad (20)$$

Since $\Lambda < N$ by Assumption 2, there exists a sufficiently small $\psi > 0$ such that

$$(1 + \psi) \frac{\Lambda^2}{N^2} < 1. \quad (21)$$

Then there exists a sufficiently large $\rho > 0$ such that

$$\frac{(N-1)^2}{N^2} \frac{\chi}{\rho} + (1 + \psi) \frac{\Lambda^2}{N^2} < 1. \quad (22)$$

Finally, there exists a sufficiently small $\phi > 0$ such that

$$\nu := \max \left\{ 1 - \frac{\lambda(Q)}{2\lambda(P)}, \frac{(N-1)^2}{N^2} \frac{\chi}{\rho} + (1 + \psi) \frac{\Lambda^2}{N^2} \right\} + \left(1 + \frac{1}{\psi} \right) \phi^2 \rho \quad (23)$$

satisfies $\nu < 1$.

B. Searching stages

Consider a sampling time t_k in a searching stage. On $[t_k, t_{k+1})$, combining (1) and (10) with $u \equiv 0$ and $\hat{x}(t_k) = x_k^*$ yields the closed-loop system

$$\dot{x} = Ax + Dd, \quad \dot{\hat{x}} = A\hat{x}, \quad (24)$$

and thus the error dynamics

$$\dot{e} = Ae + Dd, \quad e(t_k) = x(t_k) - x_k^*.$$

Hence

$$\begin{aligned} |e(t_{k+1}^-)| &\leq \|e^{A\tau_s}\| |e(t_k)| + \left(\int_0^{\tau_s} \|e^{As} D\| ds \right) \|d\|_{[t_k, t_{k+1}]} \\ &\leq \Lambda |e(t_k)| + \Phi \|d\|_{[t_k, t_{k+1}]}, \end{aligned} \quad (25)$$

where Λ and Φ are defined in (2) and (15), respectively.

We define the update formulas by

$$x_{k+1}^* = \hat{F}(x_k^*) := \hat{x}(t_{k+1}^-) = \hat{S}x_k^* \quad (26)$$

with $\hat{S} := e^{A\tau_s}$, and

$$E_{k+1} = \hat{G}((1 + \varepsilon)E_k, \delta) := (1 + \varepsilon)\Lambda E_k + \Phi\delta, \quad (27)$$

where $\varepsilon, \delta > 0$ are arbitrary design parameters introduced in Section IV-B. The functions $\hat{F}$ and $\hat{G}$ are designed such that

$$|x(t_{k+1}) - x_{k+1}^*| \leq \hat{G}(|x(t_k) - x_k^*|, \|d\|_{[t_k, t_{k+1}]}).$$

Consequently, the factor $1 + \varepsilon$ in (27) ensures that the growth of E_k eventually outpaces that of $|x(t_k) - x_k^*|$, thereby guaranteeing finite-time (re)capture of the state. The parameter δ serves as a constant disturbance estimate used only during searching stages. In contrast, [14] uses an increasing sequence of constants to estimate the disturbance during both stages, yielding only practical ISS. This distinction is one of the key ingredients for establishing ISS in the present paper.

At an escape time t_j (i.e., the start of a searching stage), the update of E_{j+1} is modified. Instead of applying (27) with $E_j = G(E_{j-1}, x_{j-1}^*)$, we define

$$E_{j+1} := \hat{G}((1 + \varepsilon)\hat{E}_j, \delta) \quad (28)$$

with

$$\hat{E}_j := \frac{\Lambda}{N} E_{j-1} + \Phi\delta.$$

This modification ensures that the state is recaptured at t_{j+1} whenever $\|d\|_{[t_{j-1}, t_{j+1}]} \leq \delta$, which is later used to bound the recapture time in terms of the disturbance.

VI. STABILITY ANALYSIS

In this section, we establish Theorem 1. We first summarize several key intermediate results as technical lemmas in Sections VI-A and VI-B; their proofs are omitted due to space limitations and can be found in [20]. Section VI-C then completes the proof of Theorem 1.

Throughout the analysis, we assume that $\|d\|_{[0, \infty)}$ is finite, since otherwise the ISS bounds (3) and (4) hold trivially.

A. Stabilizing stages

We first establish that, during a stabilizing stage, the function $V_k := V(x_k^*, E_k)$ defined in (18) is a Lyapunov-type function with exponential decay.

Lemma 1. *For any sampling time t_k in a stabilizing stage,*

$$V_{k+1} \leq \nu V_k, \quad (29)$$

where $\nu \in (0, 1)$ is defined in (23).

We now relate V_k to $|x(t_k)|$, $|x(t_{k+1})|$, and E_k .

Lemma 2. *There exist constants $C_1, C_2, C_3 > 0$ such that, for any sampling time t_k in a stabilizing stage,*

$$\sqrt{V_k} \leq C_1(|x(t_k)| + E_k), \quad (30)$$

$$|x(t_k)| \leq C_2 \sqrt{V_k}, \quad (31)$$

$$|x(t_{k+1})| \leq C_3 \sqrt{V_k} + \Phi \|d\|_{[t_k, t_{k+1}]}. \quad (32)$$

The next two lemmas provide an exponentially decaying state bound and a state bound with suitable $\mathcal{K}_\infty$ properties.

Lemma 3. *There exist constants $C, \lambda > 0$ such that, for any two sampling times $t_k > t_l$ from the same stabilizing stage,*

$$|x(t_k)| \leq C e^{-\lambda(k-l)} (|x(t_l)| + E_l) + \Phi \|d\|_{[t_k, t_l]}. \quad (33)$$

Lemma 4. *There exist continuous functions $\chi^x, \chi^d : \mathbb{R}_{>0} \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ such that, a) for each fixed $E > 0$, $\chi^x(E, \cdot), \chi^d(E, \cdot) \in \mathcal{K}_\infty$, b) for each fixed $s > 0$, $\chi^x(\cdot, s)$ and $\chi^d(\cdot, s)$ are nondecreasing, and c) for any two sampling times $t_k \geq t_l$ from the same stabilizing stage,*

$$|x(t_k)| \leq \chi^x(E_l, |x(t_l)|) + \chi^d(E_l, \|d\|_{[t_l, t_k]}). \quad (34)$$

Finally, we bound the state and the quantization radius at escape times in terms of the disturbance over the preceding sampling interval, independently of the initial state.

Lemma 5. *There exists a constant $\Gamma > 0$ such that, for any sampling time t_j at which the state escapes,*

$$|x(t_j)| \leq \Gamma \|d\|_{[t_{j-1}, t_j]}, \quad E_{j-1} \leq \Gamma \|d\|_{[t_{j-1}, t_j]}. \quad (35)$$

B. Searching stages

We first establish that if the state is lost at $t_0 = 0$, then it is guaranteed to be captured in finite time.

Lemma 6. *If the state is lost at $t_0 = 0$, then it is captured at some sampling time t_{i_0} satisfying*

$$i_0 \leq \max\left\{\eta_x(|x_0|/E_0), \eta_d(\|d\|_{[0,t_{i_0}]}/\delta)\right\} \quad (36)$$

with $\eta_x(s) := \lceil \log_{1+\varepsilon} s \rceil$ and $\eta_d(s) := \lceil \log_{1+\varepsilon}(r_\varepsilon s) \rceil$ for $s > 1$, and $\eta_x(s) = \eta_d(s) = 0$ for $0 \leq s \leq 1$, where $r_\varepsilon := (\hat{\Lambda} - 1)/(\Lambda - 1)$ and $\hat{\Lambda} := (1 + \varepsilon)\Lambda$.

The next lemma provides bounds on the state before the first capture and on the quantization radius at the first capture.

Lemma 7. *Suppose that the state is first captured at t_{i_0} . There exist continuous functions $\hat{\gamma}_0^x, \hat{\gamma}_0^d : \mathbb{R}_{>0} \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ such that, a) for each fixed $E > 0$, $\hat{\gamma}_0^x(E, \cdot), \hat{\gamma}_0^d(E, \cdot) \in \mathcal{K}_\infty$, and b) for any sampling time $t_k \leq t_{i_0}$,*

$$|x(t_k)| \leq \hat{\gamma}_0^x(E_0, |x_0|) + \hat{\gamma}_0^d(E_0, \|d\|_{[0,t_{i_0}]}). \quad (37)$$

Moreover, there exists a continuous function $\hat{\chi}_0^E : \mathbb{R}_{>0} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ such that, a) for each fixed $E > 0$ and $s \geq 0$, $\hat{\chi}_0^E(E, \cdot, s)$ and $\hat{\chi}_0^E(E, s, \cdot)$ are nondecreasing, and b) at the first capture time t_{i_0} ,

$$E_{i_0} \leq \hat{\chi}_0^E(E_0, |x_0|, \|d\|_{[0,t_{i_0}]}). \quad (38)$$

We now establish that if the state escapes, then it is guaranteed to be recaptured in finite time.

Lemma 8. *If the state escapes at $t_j > 0$, then it is recaptured at some sampling time $t_i > t_j$ satisfying*

$$i \leq j + \max\{\eta_d(\|d\|_{[t_{j-1}, t_i]}/\delta), 1\}, \quad (39)$$

where η_d is defined in Lemma 6.

The final lemma provides bounds on the state before recapture and on the quantization radius at recapture.

Lemma 9. *Suppose that the state escapes at t_j and is recaptured at t_i . There exist $\hat{\gamma}^x, \hat{\gamma}^d \in \mathcal{K}_\infty$ such that, for any sampling time t_k between t_j and t_i ,*

$$|x(t_k)| \leq \hat{\gamma}^x(|x(t_j)|) + \hat{\gamma}^d(\|d\|_{[t_{j-1}, t_i]}). \quad (40)$$

Moreover, there exists a continuous function $\hat{\chi}^E : \mathbb{R}_{>0} \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ such that a) for each fixed $E > 0$ and $s \geq 0$, $\hat{\chi}^E(E, \cdot)$ and $\hat{\chi}^E(\cdot, s)$ are nondecreasing, and b) at the recapture time t_i ,

$$E_i \leq \hat{\chi}^E(E_{j-1}, \|d\|_{[t_{j-1}, t_i]}). \quad (41)$$

C. Proof of Theorem 1

The proof proceeds by alternating between searching and stabilizing stages. During searching stages, Lemmas 6 and 8 ensure finite-time (re)capture, while Lemmas 7 and 9 provide bounds on the state and the quantization radius. During stabilizing stages, Lemmas 3 and 4 provide state bounds, while Lemma 5 bounds the state and the quantization radius

at escape times in terms of the disturbance. Combining these bounds yields the ISS bounds (3) and (4).

We index the alternating searching and stabilizing stages as follows. Let $0 \leq i_0 < j_1 < i_1 < \dots$ be such that the state is first captured at t_{i_0} , and for each $l \geq 1$, it escapes at t_{j_l} and is recaptured at t_{i_l} . By Lemma 6, we have $i_0 < \infty$. If the state never escapes after some t_{i_l} , we set $j_{l+1} = \infty$. By Lemma 8, if $j_l < \infty$, then $i_l < \infty$.

First, we establish the bound (3) at sampling times by constructing the functions $\gamma_1, \gamma_2 \in \mathcal{K}_\infty$.

First searching stage $[0, t_{i_0}]$: By Lemma 7, the bound (37) holds for all $t_k \leq t_{i_0}$, including the case $i_0 = 0$ (i.e., when the state is visible at $t_0 = 0$).

First stabilizing stage $[t_{i_0}, t_{j_1}]$: At the first capture time t_{i_0} , substituting the bounds (37) and (38) from Lemma 7 into (34) from Lemma 4 gives, for any $t_{i_0} \leq t_k < t_{j_1}$,

$$\begin{aligned} |x(t_k)| &\leq \chi^x\left(\hat{\chi}_0^E(E_0, |x_0|, \|d\|_{[0,t_{i_0}]}), \right. \\ &\quad \left. \hat{\gamma}_0^x(E_0, |x_0|) + \hat{\gamma}_0^d(E_0, \|d\|_{[0,t_{i_0}]})\right) \\ &\quad + \chi^d\left(\hat{\chi}_0^E(E_0, |x_0|, \|d\|_{[0,t_{i_0}]}), \|d\|_{[t_{i_0}, t_k]}\right). \end{aligned}$$

Therefore, by a standard comparison-function construction, there exist continuous functions $\bar{\chi}_1, \bar{\chi}_2 : \mathbb{R}_{>0} \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ such that a) for each fixed $E > 0$, $\bar{\chi}_1(E, \cdot), \bar{\chi}_2(E, \cdot) \in \mathcal{K}_\infty$, and b) for any $t_{i_0} \leq t_k < t_{j_1}$,

$$|x(t_k)| \leq \bar{\chi}_1(E_0, |x_0|) + \bar{\chi}_2(E_0, \|d\|_{[0,t_k]}).$$

If $t_{j_1} = \infty$, the proof of (3) at sampling times is complete. Otherwise, consider an arbitrary escape time $t_{j_l} < \infty$.

Searching stage $[t_{j_l}, t_{i_l}]$: At the escape time t_{j_l} , substituting the first bound in (35) from Lemma 5 into (40) from Lemma 9 gives, for any $t_{j_l} \leq t_k \leq t_{i_l}$,

$$|x(t_k)| \leq \hat{\gamma}(\|d\|_{[t_{j_l-1}, t_{i_l}]}),$$

where $\hat{\gamma}(s) := \hat{\gamma}^x(\Gamma s) + \hat{\gamma}^d(s) \in \mathcal{K}_\infty$.

Stabilizing stage $[t_{i_l}, t_{j_{l+1}}]$: At the recapture time t_{i_l} ,

$$|x(t_{i_l})| \leq \hat{\gamma}(\|d\|_{[t_{j_l-1}, t_{i_l}]}),$$

while the second bound in (35) from Lemma 5 and (41) from Lemma 9 yield

$$E_{i_l} \leq \hat{\chi}^E(\Gamma \|d\|_{[t_{j_l-1}, t_{j_l}]}, \|d\|_{[t_{j_l-1}, t_{i_l}]}).$$

Substituting these bounds into (34) from Lemma 4 gives, for any $t_{i_l} \leq t_k < t_{j_{l+1}}$,

$$|x(t_k)| \leq \bar{\gamma}(\|d\|_{[t_{j_l-1}, t_k]}),$$

with $\bar{\gamma}(s) := \chi^x(\hat{\chi}^E(\Gamma s, s), \hat{\gamma}(s)) + \chi^d(\hat{\chi}^E(\Gamma s, s), s) \in \mathcal{K}_\infty$.

Combining the above cases yields the bound (3) for all sampling times with the $\mathcal{K}_\infty$ functions

$$\begin{aligned} \gamma_1(s) &:= \max\{\hat{\gamma}_0^x(E_0, s), \bar{\chi}_1(E_0, s)\}, \\ \gamma_2(s) &:= \max\{\hat{\gamma}_0^d(E_0, s), \bar{\chi}_2(E_0, s), \hat{\gamma}(s), \bar{\gamma}(s)\}. \end{aligned} \quad (42)$$

Next, we establish the bound (4) at sampling times by constructing the function $\gamma_3 \in \mathcal{K}_\infty$. If only finitely many searching stages occur, then the system eventually

remains in a stabilizing stage, and Lemma 3 implies $\limsup_{k \rightarrow \infty} |x(t_k)| \leq \Phi \limsup_{k \rightarrow \infty} \|d\|_{[t_{k-1}, t_k]}$. If infinitely many searching stages occur, then the bounds constructed in the above derivation of (3) imply that, between escape times t_{j_l} and $t_{j_{l+1}}$, we have $|x(t_k)| \leq \hat{\gamma}(\|d\|_{[t_{j_l-1}, t_{j_l}]})$ for all $k \in \{j_l, \dots, i_l\}$, and $|x(t_k)| \leq \bar{\gamma}(\|d\|_{[t_{j_{l+1}-1}, t_{j_{l+1}}]})$ for all $k \in \{i_l, \dots, j_{l+1} - 1\}$. Therefore, $\limsup_{k \rightarrow \infty} |x(t_k)| \leq \gamma_3(\limsup_{t \rightarrow \infty} |d(t)|)$ with the $\mathcal{K}_\infty$ function

$$\gamma_3(s) := \max\{\Phi s, \hat{\gamma}(s), \bar{\gamma}(s)\}. \quad (43)$$

The extension of (3) and (4) from sampling times to all $t \geq 0$ follows from standard arguments.

VII. SIMULATION EXAMPLE

We illustrate the proposed communication and control strategy on the linear system (1) with

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1.5 \end{bmatrix}, B = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}, D = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, K = \begin{bmatrix} -3.5 & 0 \end{bmatrix},$$

sampling period $\tau_s = 0.1$ s, and integer $N = 5$, which satisfy Assumptions 1 and 2. The design parameters are selected as described in Sections IV and V:

$$E_0 = 0.5, \varepsilon = 0.2, \delta = 0.1, \psi = 0.5, \rho = 150, \phi = 0.01.$$

The simulation starts from the initial state $x_0 = [1 \ 1]^\top$, which lies outside the initial quantization range. To induce escape and recapture events, disturbance pulses of magnitude 1.5 and duration 0.2 s are applied at $t = 3$ s, 9.5 s, and 11.2 s; otherwise, $d(t) = 0$.

Figure 2(a) illustrates the escape and recapture mechanism. During each searching stage, E_k eventually grows faster than $|e(t_k)|$, ensuring finite-time (re)capture. After (re)capture, both $|e(t_k)|$ and E_k decrease rapidly at sampling times during the stabilizing stage, although $|e(t)|$ may increase between sampling times. Disturbance pulses cause subsequent escapes and initiate new searching stages.

Figure 2(b) illustrates the corresponding evolution of the first components of the system and auxiliary states. During searching stages, the control input is set to zero, and $x_1(t)$ and $\hat{x}_1(t)$ grow in the absence of stabilizing control. After (re)capture, the control input $u(t) = -3.5\hat{x}_1(t)$ drives them toward zero. The auxiliary state evolves continuously between sampling times and jumps when it is updated using a new quantized measurement. These behaviors are consistent with the ISS properties established in Theorem 1.

VIII. CONCLUSION

This paper studied feedback stabilization of linear systems under data-rate constraints in the presence of completely unknown disturbances. Using dynamic quantization and a new disturbance-estimation design, the proposed approach establishes input-to-state stability (ISS) with respect to the disturbance. Future work will focus on extensions to switched and nonlinear dynamics and applications to networked control systems.

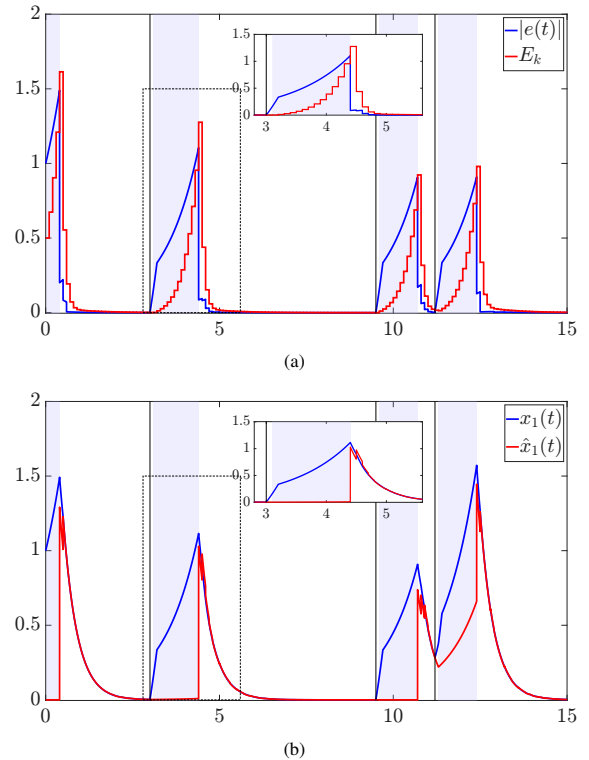


Fig. 2: Evolution of (a) the error $|e(t)| = |x(t) - \hat{x}(t)|$ and the quantization radius E_k , and (b) the first components $x_1(t)$ and $\hat{x}_1(t)$ of the system and auxiliary states. Black vertical lines indicate disturbance onsets. Shaded regions indicate searching stages. The zoomed-in plots highlight the first escape and the subsequent recapture.

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